

OOPS ASSIGNMENT – 03

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Batch – 2023

1.

```
#include <iostream>
```

```
using namespace std;
```

```
class Student {
```

```
private:
```

```
    int roll;
```

```
    float marks;
```

```
public:
```

```
    void setData(int r, float m) {
```

```
        this->roll = r;
```

```
        this->marks = m;
```

```
    }
```

```
    void display() {
```

```
        cout << "Roll: " << roll << endl;
```

```
        cout << "Marks: " << marks << endl;
```

```
    }
```

```
};
```

```

int main() {

    Student s1;

    s1.setData(101, 92.5);

    s1.display();


    Student *ptr;

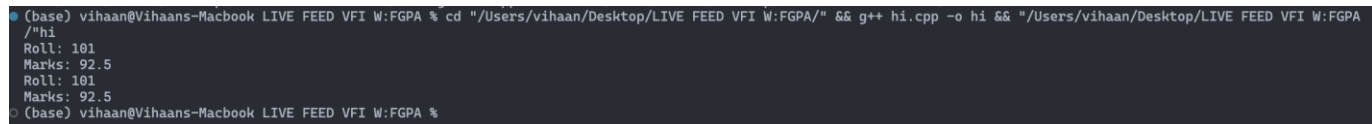
    ptr = &s1;

    ptr->display();

    return 0;

}

```



```

(base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA % cd "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/" && g++ hi.cpp -o hi && "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA
/hi
Roll: 101
Marks: 92.5
Roll: 101
Marks: 92.5
(base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA %

```

2.

```

#include <iostream>

using namespace std;

```

```

class B;

```

```

class A {
private:
    int x;
public:
    A(int a) {
        x = a;
    }
}

```

```
    }  
    friend void swapValues(A &, B &);  
    void display() {  
        cout << "A: " << x << endl;  
    }  
};
```

```
class B {  
private:  
    int y;  
public:  
    B(int b) {  
        y = b;  
    }  
    friend void swapValues(A &, B &);  
    void display() {  
        cout << "B: " << y << endl;  
    }  
};
```

```
void swapValues(A &obj1, B &obj2) {  
    int temp = obj1.x;  
    obj1.x = obj2.y;  
    obj2.y = temp;  
}
```

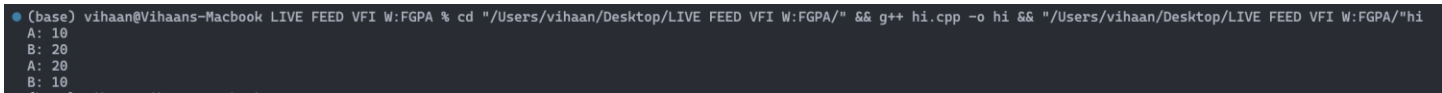
```
int main() {  
    A obj1(10);  
    B obj2(20);
```

```
obj1.display();  
obj2.display();  
  
swapValues(obj1, obj2);
```

```
obj1.display();  
obj2.display();
```

```
return 0;
```

```
}
```



```
(base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA % cd "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/" && g++ hi.cpp -o hi && "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/"hi  
A: 10  
B: 20  
A: 20  
B: 10
```

3.

```
#include <iostream>  
using namespace std;
```

```
class B;
```

```
class A {
```

```
private:
```

```
    int x;
```

```
public:
```

```
    A(int a) {
```

```
        x = a;
```

```
    }
```

```
    friend int add(A, B);
```

```
};
```

```
class B {
```

```
private:
```

```

    int y;
public:
    B(int b) {
        y = b;
    }

    friend int add(A, B);
};

int add(A obj1, B obj2) {
    return obj1.x + obj2.y;
}

int main() {
    A obj1(10);
    B obj2(20);

    int result = add(obj1, obj2);

    cout << "Sum: " << result << endl;

    return 0;
}

```

```

● (base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA % cd "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/" && g++ hi.cpp -o hi && "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/"hi
Sum: 30
○ (base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA %

```

4.

```

#include <iostream>

using namespace std;

```

```

class B;

```

```

class A {

```

private:

int x;

public:

A(int a) {

x = a;

}

friend class B;

};

class B {

public:

void show(A obj) {

cout << "Value of x: " << obj.x << endl;

}

};

int main() {

A obj(50);

B b;

b.show(obj);

return 0;

}

```
• (base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA % cd "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/" && g++ hi.cpp -o hi && "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/"hi
Value of x: 50
○ (base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA %
```

5.

#include <iostream>

using namespace std;

class Rectangle {

private:

```
    int length, breadth;

public:
    void setData(int l, int b) {
        length = l;
        breadth = b;
    }
    int area() {
        return length * breadth;
    }
};

int main() {
    int n;
    cin >> n;

    Rectangle r[n];

    for(int i = 0; i < n; i++) {
        int l, b;
        cin >> l >> b;
        r[i].setData(l, b);
    }

    for(int i = 0; i < n; i++) {
        cout << "Area of rectangle " << i + 1 << ": " << r[i].area() << endl;
    }

    return 0;
}
```

```

● cd "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/" && g++ hi.cpp -o hi && "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/"hi
Area of rectangle 1: 30
Area of rectangle 2: 56
Area of rectangle 3: 0
Area of rectangle 4: 0
○ (base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA % █

```

6.

```
#include using namespace std;
```

```
inline int cube(int x) { return x * x * x; }
```

```
int main() { int n; cin >> n; cout << cube(n) << endl; return 0; }
```

```

● (base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA % cd "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/" && g++ hi.cpp -o hi && "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/"hi
5
125
○ (base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA % █

```

7. (a)

```
#include <iostream>
```

```
using namespace std;
```

```
class Number {
```

```
public:
```

```
    int x;
```

```
    Number(int a) {
```

```
        x = a;
```

```
    }
```

```
};
```

```
Number modify(Number obj) {
```

```
    obj.x = obj.x + 10;
```

```
    return obj;
```

```
}
```

```
int main() {
```

```
    Number n1(5);
```



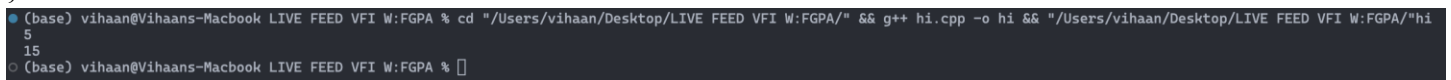
```
Number n2 = modify(n1);
```

```
cout << n1.x << endl;
```

```
cout << n2.x << endl;
```

```
return 0;
```

```
}
```

A terminal window on a Mac showing the compilation and execution of a C++ program. The prompt is (base) vihaan@Vihaans-Macbook. The command is cd "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/" && g++ hi.cpp -o hi && "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/"hi. The output is 5 and 15 on two separate lines. The prompt is then (base) vihaan@Vihaans-Macbook.

```
(base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA % cd "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/" && g++ hi.cpp -o hi && "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/"hi
5
15
(base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA %
```

(b).

```
#include <iostream>
```

```
using namespace std;
```

```
class Number {
```

```
public:
```

```
int x;
```

```
Number(int a) {
```

```
    x = a;
```

```
}
```

```
};
```

```
Number modify(Number *obj) {
```

```
    obj->x = obj->x + 10;
```

```
    return *obj;
```

```
}
```

```
int main() {
```

```
    Number n1(5);
```

```
    Number n2 = modify(&n1);
```

```
cout << n1.x << endl;
```

```
cout << n2.x << endl;
```

```
return 0;
```

```
}
```

```
● (base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA % cd "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/" && g++ hi.cpp -o hi && "/Users/vihaan/Desktop/LIVE FEED VFI W:FGPA/"hi
15
15
○ (base) vihaan@Vihaans-Macbook LIVE FEED VFI W:FGPA %
```